

Bike Station Performance Dashboard

1. Project Overview & Objective

Project Overview

This project analyzes bike station availability, station capacity, banking facilities, bonus stations, and operational status using **Power BI**. The objective is to convert raw bike station data into meaningful insights through interactive dashboards to support city transportation management.

Objective

- Analyze bike availability across stations.
 - Monitor station capacity and utilization.
 - Evaluate Banking and Bonus station distribution.
 - Track station operational status (Open/Closed).
 - Build an interactive Power BI dashboard for operational decision-making.
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2. Data Source

Source

- JCDecaux Bike Station Dataset
- Domain: Transportation / Smart Mobility

Dataset Includes

- Station Number
 - Station Name
 - Address
 - Position
 - Banking
 - Bonus
 - Status
 - Contract Name
 - Bike Stands
 - Available Bike Stands
 - Available Bikes
 - Date & Time Last Update
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3. Problem Statement

The objective is to monitor bike station performance by analyzing station availability, bike capacity, operational status, banking support, and bonus facilities. The dashboard helps identify high-demand stations and improve bike redistribution and maintenance.

4. Attribute Details

Column	Description
Number	Station ID
Name	Station Name
Address	Station Address
Position	Latitude & Longitude
Banking	Banking Facility
Bonus	Bonus Station
Status	Open / Closed
Contract Name	City
Bike Stands	Total Capacity
Available Bike Stands	Empty Bike Stands
Available Bikes	Bikes Available
Date & Time Last Update	Last Updated Time

5. Tools & Technologies

- Microsoft Excel
 - Power Query
 - Power BI
 - DAX
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6. Data Preprocessing

Tasks Performed

- Removed duplicate records.
- Removed unnecessary columns.
- Standardized data types.
- Cleaned station names.
- Removed station ID from Name column.
- Trimmed unwanted spaces.
- Split Position into Latitude and Longitude (if used in Map).
- Imported cleaned data into Power BI.

7. Data Modeling & DAX

Data Model

Fact Table

- Bike Station Data

Dimension Tables (Optional)

- Status
- Banking
- Bonus
- Contract Name

Relationship

- Cross Filter Direction: Single
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DAX Measuree

```
Bike Availability % = DIVIDE(SUM(Feuil1[Available Bikes]),SUM(Feuil1[Bike Stands]),0)
Station status = SWITCH(TRUE(),
Feuil1[Status]="OPEN", "Active",
Feuil1[Status]="CLOSED", "Inactive", "Other")
Total Available Bike = SUM(Feuil1[Available Bikes])
Total Bike Stands = SUM(Feuil1[Available Bike Stands])
Total Bikes = SUM(Feuil1[Bike Stands])
Total station = COUNT(Feuil1[Bike Stands])
```

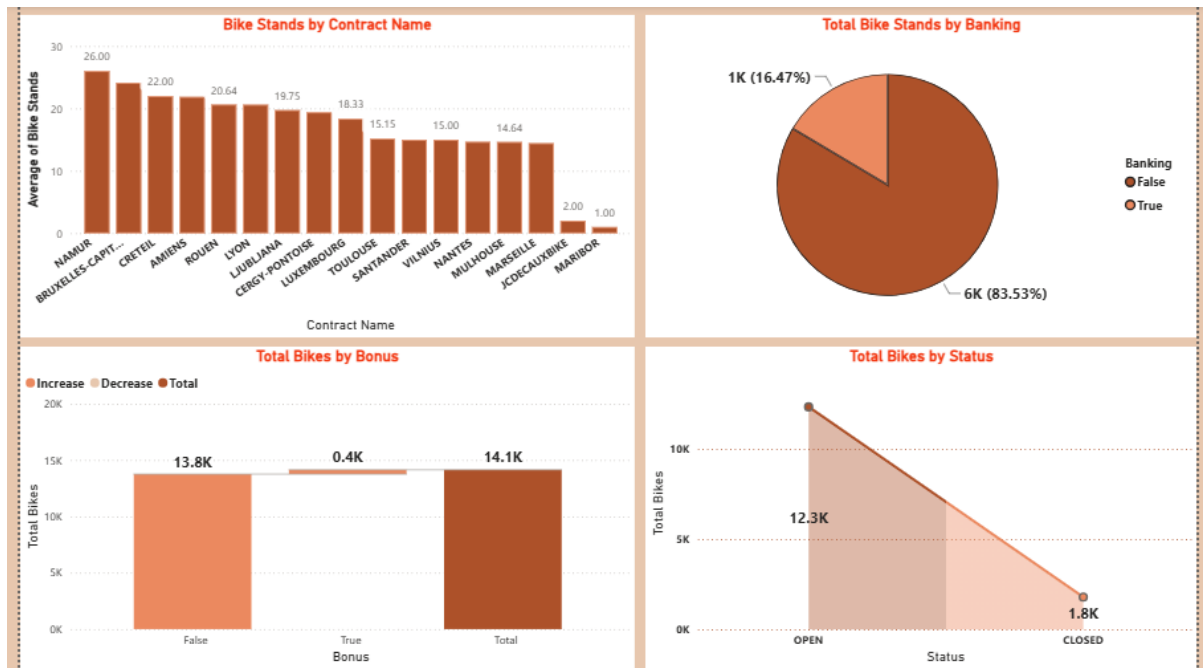
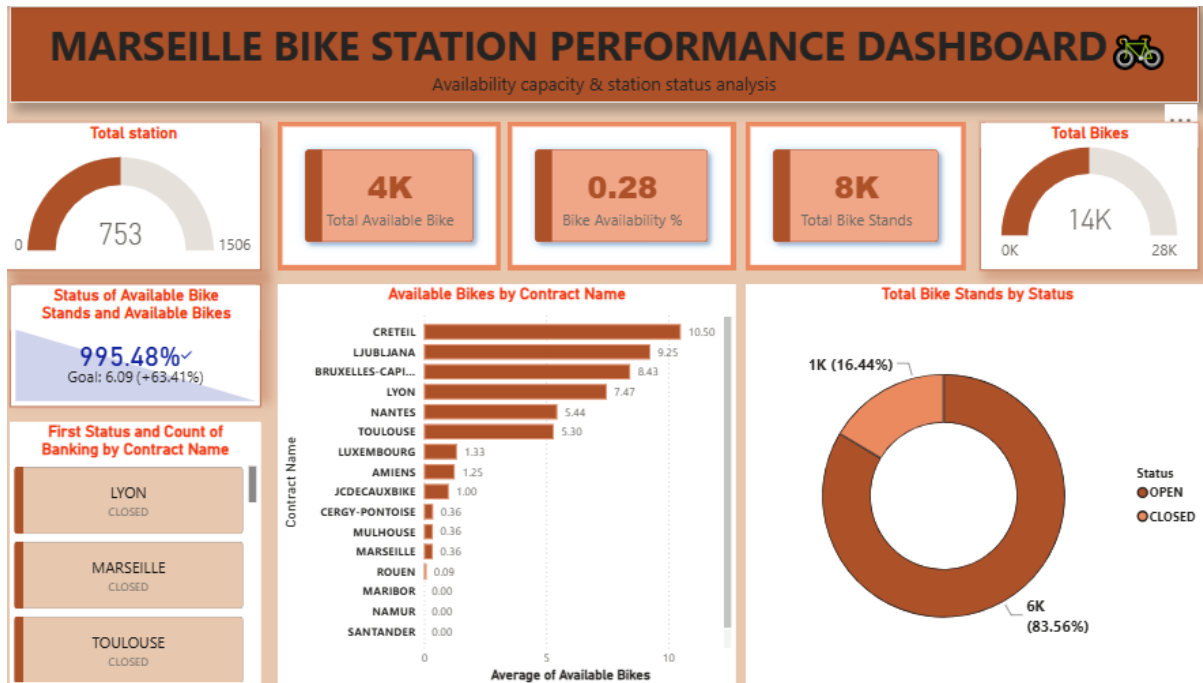
8. Analysis & Visualizations

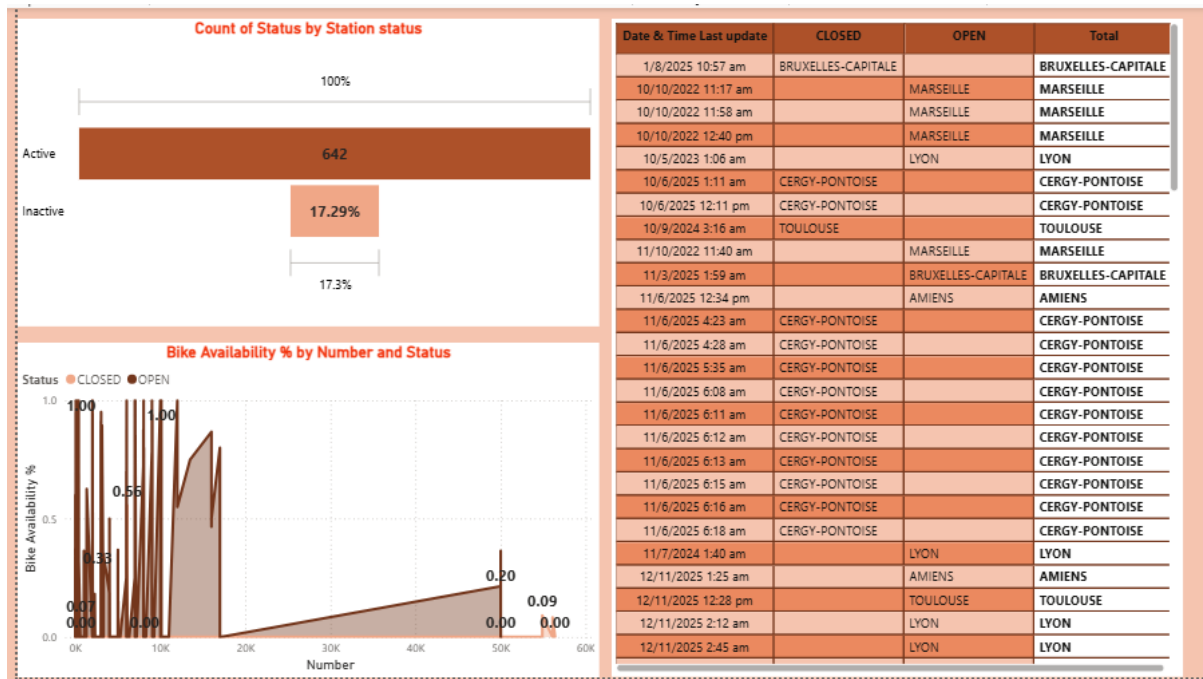
KPI Cards

- Total Stations
- Total Bike Stands
- Available Bikes
- Available Bike Stands
- Open Stations
- Banking Stations
- Bonus Stations

- Bike Availability %

Charts





Slicers

- Contract Name
- Status
- Banking
- Bonus

Interactive Features

- Drill Down
- Cross Filtering
- Slicers
- Tooltips
- Data Labels

9. Key Insights

- Total bike stations and overall capacity can be monitored through KPI cards.
- Stations with the highest available bikes indicate lower demand at that moment.
- Stations with the lowest available bikes indicate higher usage and may require bike redistribution.
- Most stations are operational (OPEN), ensuring better service availability.

- Banking-enabled stations form a major portion of the network.
 - Bonus stations encourage improved bike circulation but represent a smaller share of the network.
 - The time trend shows how bike availability changes based on the latest updates.
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10. Recommendations

- Increase bike redistribution to stations with consistently low bike availability.
 - Monitor high-demand stations more frequently.
 - Expand the number of Bonus stations to improve bike circulation.
 - Perform regular maintenance on stations with recurring operational issues.
 - Use the dashboard to support real-time monitoring and operational planning.
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11. Conclusion

This project demonstrates how Power BI can transform raw bike station data into meaningful operational insights. By using Power Query for data cleaning, DAX for calculations, and interactive dashboards for visualization, stakeholders can monitor bike availability, station capacity, and operational performance to make informed decisions and improve urban mobility.